Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/02/2004
PI Name:	Riley, James J. Ph.D.		
Project Title:	Investigation of the Liftoff and Blowout of Transitional and Turbulent Jet Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	99-HEDS-04
Start Date:	01/09/2001	**End Date:**	07/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2517		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	We are applying three-dimensional direct numerical simulation and concurrent theoretical analysis to study turbulent jet flame liftoff and blowout for a variety of buoyancy conditions. There are competing theories regarding jet flame liftoff and blowout, and especially how this is influenced by buoyancy. Our objectives are to use numerical simulations and theoretical analysis to (i) test existing hypotheses, (ii) identify the governing mechanisms for liftoff and blowout, (iii) determine which of the various modeling approaches are most consistent with the governing mechanisms, and (iv) devise a means for controlling liftoff and blowout. Turbulent jet flame liftoff and blowout are of critical importance to the safety and reliability of a variety of combustion devices. Our research will help understand these phenomena, and better predict them. This information can then be used in the design of safer and more reliable combustion devices.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	
Task Progress:	Our research objective is to study the process of liftoff and lifted flames in microgravity environments. Specifically, we are interested in the effects of buoyancy and forcing upon turbulent and transitioning lifted flames. During the past year we have completed the development of our computer code for simulating buoyant and non-buoyant jet flames. The code has been developed to run on massively parallel computers, and employs state-of-the-art as well as newly developed numerical methods. This year we have applied this code to a study of the three-dimensional transitioning region of buoyant and non-buoyant jet flames. The simulations clearly show how the jet flame is strongly affected by buoyancy. We have also performed linear stability analysis, using a primitive variable formulation of the linearized Navier-Stokes equations. From this analysis it was determined that, below a certain threshold density ratio, the flows becomes absolutely unstable, in that it then supports sustained oscillations. Furthermore the Strouhal number predicted by the linearized theory correlates well with that observed in the direct numerical simulations. These theoretical and numerical results suggest optimal disturbance forcing of the jet flame in order to optimize mixing. We are presently attempting to exploit these results in further direct numerical simulations.
Bibliography Type:	Description: (Last Updated: 11/02/2004)
Abstracts for Journals and Proceedings	Nichols, J.W.; Schmid, P.J.; Riley, J.J "Self-sustained oscillations in variable-density jets" Bulletin of the American Physical Society Not yet available , Dec-2004